

ISLAMIC UNIVERSITY OF TECHNOLOGY



DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING

CSE 4631

DIGITAL SIGNAL PROCESSING

Class Test 3 & 4 Solutions

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1 Question 1: Discrete-Time Signals and Systems

1.1 (a) Signal Manipulations

Given

$$x[n] = \{2, -1, 3, 4, -2\}$$

where

$$x[0] = 3.$$

Therefore, the index locations of the samples are

$$\begin{aligned}x[-2] &= 2, \\x[-1] &= -1, \\x[0] &= 3, \\x[1] &= 4, \\x[2] &= -2.\end{aligned}$$

Hence,

$$\boxed{x[n] = \{2, -1, \underset{\uparrow}{3}, 4, -2\}}$$

where the arrow indicates the sample at $n = 0$.

1.1.1 i. $x[n]$

The original signal is

$$\boxed{x[n] = \{2, -1, \underset{\uparrow}{3}, 4, -2\}}$$

with

$$x[-2] = 2, \quad x[-1] = -1, \quad x[0] = 3, \quad x[1] = 4, \quad x[2] = -2.$$

1.1.2 ii. $x[n-2]$

The expression $x[n-2]$ represents a shift of the signal two units to the right.

For the shifted signal,

$$\begin{aligned}x[n-2] &= x[-2] = 2 && \text{at } n = 0, \\x[n-2] &= x[-1] = -1 && \text{at } n = 1, \\x[n-2] &= x[0] = 3 && \text{at } n = 2, \\x[n-2] &= x[1] = 4 && \text{at } n = 3, \\x[n-2] &= x[2] = -2 && \text{at } n = 4.\end{aligned}$$

Therefore,

$$x[n-2] = \{2, -1, 3, 4, -2\}$$

↑

1.1.3 iii. $x[-n]$

The expression $x[-n]$ reverses the signal around $n = 0$.

The samples become

$$\begin{aligned}x[-(-2)] &= x[2] = -2 && \text{at } n = -2, \\x[-(-1)] &= x[1] = 4 && \text{at } n = -1, \\x[-0] &= x[0] = 3 && \text{at } n = 0, \\x[-1] &= -1 && \text{at } n = 1, \\x[-2] &= 2 && \text{at } n = 2.\end{aligned}$$

Therefore,

$$x[-n] = \{-2, 4, 3, -1, 2\}$$

↑

1.1.4 iv. $x[2-n]$

We can write

$$x[2-n] = x[-(n-2)].$$

This corresponds to time reversal followed by a shift.

The resulting samples are

$$\begin{aligned}
x[2-0] &= x[2] = -2 && \text{at } n = 0, \\
x[2-1] &= x[1] = 4 && \text{at } n = 1, \\
x[2-2] &= x[0] = 3 && \text{at } n = 2, \\
x[2-3] &= x[-1] = -1 && \text{at } n = 3, \\
x[2-4] &= x[-2] = 2 && \text{at } n = 4.
\end{aligned}$$

Therefore,

$$x[2-n] = \{\underset{\uparrow}{-2}, 4, 3, -1, 2\}$$

1.2 (b) System Properties

The given system is

$$y[n] = nx[n] + x[n-1].$$

1.2.1 Linearity

Consider two inputs $x_1[n]$ and $x_2[n]$ with outputs

$$y_1[n] = nx_1[n] + x_1[n-1]$$

and

$$y_2[n] = nx_2[n] + x_2[n-1].$$

Let

$$x_3[n] = ax_1[n] + bx_2[n].$$

Then,

$$\begin{aligned}
y_3[n] &= nx_3[n] + x_3[n-1] \\
&= n(ax_1[n] + bx_2[n]) + ax_1[n-1] + bx_2[n-1] \\
&= a(nx_1[n] + x_1[n-1]) + b(nx_2[n] + x_2[n-1]) \\
&= ay_1[n] + by_2[n].
\end{aligned}$$

Therefore,

The system is linear.

1.2.2 Time Invariance

For the original input $x[n]$,

$$y[n] = nx[n] + x[n - 1].$$

Now consider a shifted input

$$x_k[n] = x[n - k].$$

The output corresponding to this shifted input is

$$y_k[n] = nx[n - k] + x[n - k - 1].$$

On the other hand, shifting the original output gives

$$y[n - k] = (n - k)x[n - k] + x[n - k - 1].$$

Since, in general,

$$nx[n - k] \neq (n - k)x[n - k],$$

we have

$$y_k[n] \neq y[n - k].$$

Therefore,

The system is time-variant.

1.2.3 Causality

The output is

$$y[n] = nx[n] + x[n - 1].$$

It depends only on the present input $x[n]$ and the previous input $x[n - 1]$.
It does not depend on any future input sample.

Therefore,

The system is causal.

1.2.4 Stability

Assume that the input is bounded:

$$|x[n]| \leq M_x < \infty.$$

Then,

$$\begin{aligned} |y[n]| &= |nx[n] + x[n-1]| \\ &\leq |n||x[n]| + |x[n-1]| \\ &\leq |n|M_x + M_x \\ &= (|n| + 1)M_x. \end{aligned}$$

As $|n| \rightarrow \infty$, the quantity $(|n| + 1)M_x$ becomes unbounded. Therefore,

The system is unstable.

2 Question 2: Linear Convolution

Given

$$x[n] = \{1, 2, 1\}$$

and

$$h[n] = \{1, -1, 2\},$$

where the first element of each sequence corresponds to $n = 0$.
The linear convolution is defined as

$$y[n] = x[n] * h[n]$$

or

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

Since both sequences have three samples, the output length is

$$N_y = N_x + N_h - 1.$$

Thus,

$$N_y = 3 + 3 - 1 = 5.$$

Therefore, the output samples are defined for

$$n = 0, 1, 2, 3, 4.$$

2.1 (a) Mathematical Analytical Calculation

For $n = 0$,

$$\begin{aligned} y[0] &= x[0]h[0] \\ &= (1)(1) \\ &= 1. \end{aligned}$$

For $n = 1$,

$$\begin{aligned}
y[1] &= x[0]h[1] + x[1]h[0] \\
&= (1)(-1) + (2)(1) \\
&= -1 + 2 \\
&= 1.
\end{aligned}$$

For $n = 2$,

$$\begin{aligned}
y[2] &= x[0]h[2] + x[1]h[1] + x[2]h[0] \\
&= (1)(2) + (2)(-1) + (1)(1) \\
&= 2 - 2 + 1 \\
&= 1.
\end{aligned}$$

For $n = 3$,

$$\begin{aligned}
y[3] &= x[1]h[2] + x[2]h[1] \\
&= (2)(2) + (1)(-1) \\
&= 4 - 1 \\
&= 3.
\end{aligned}$$

For $n = 4$,

$$\begin{aligned}
y[4] &= x[2]h[2] \\
&= (1)(2) \\
&= 2.
\end{aligned}$$

Therefore,

$$y[n] = \{1, 1, 1, 3, 2\}$$

where the first element corresponds to $n = 0$.

2.2 (b) Tabular Verification

The products of the samples of $x[n]$ and $h[n]$ are shown below.

Table 1: Tabular Convolution			
$x \backslash h$	$h[0] = 1$	$h[1] = -1$	$h[2] = 2$
$x[0] = 1$	1	-1	2
$x[1] = 2$	2	-2	4
$x[2] = 1$	1	-1	2

Summing the appropriate terms along the convolution diagonals,

$$y[0] = 1$$

$$y[1] = 2 + (-1) = 1$$

$$y[2] = 1 + (-2) + 2 = 1$$

$$y[3] = (-1) + 4 = 3$$

$$y[4] = 2.$$

Hence,

$$y[n] = \{1, 1, 1, 3, 2\}$$

which agrees with the analytical calculation.

3 Question 3: Difference Equation and LTI System

The given LTI discrete-time system is

$$y[n] - 0.5y[n-1] = x[n] + x[n-1]$$

with initial rest conditions.

3.1 (a) Impulse Response $h[n]$

Taking the Z-transform of both sides under initial rest conditions,

$$Y(z) - 0.5z^{-1}Y(z) = X(z) + z^{-1}X(z).$$

Factoring,

$$Y(z)(1 - 0.5z^{-1}) = X(z)(1 + z^{-1}).$$

Therefore,

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1}}{1 - 0.5z^{-1}}.$$

Separating the expression,

$$H(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{z^{-1}}{1 - 0.5z^{-1}}.$$

Using the standard Z-transform pair

$$\frac{1}{1 - az^{-1}} \longleftrightarrow a^n u[n],$$

we obtain

$$\mathcal{Z}^{-1} \left\{ \frac{1}{1 - 0.5z^{-1}} \right\} = (0.5)^n u[n].$$

Also,

$$\mathcal{Z}^{-1} \left\{ \frac{z^{-1}}{1 - 0.5z^{-1}} \right\} = (0.5)^{n-1} u[n-1].$$

Therefore,

$$h[n] = (0.5)^n u[n] + (0.5)^{n-1} u[n-1]$$

3.2 (b) Output for $x[n] = u[n]$

For

$$x[n] = u[n],$$

the Z-transform is

$$X(z) = \frac{1}{1 - z^{-1}}.$$

Therefore,

$$Y(z) = H(z)X(z).$$

Substituting the expressions,

$$Y(z) = \frac{1 + z^{-1}}{(1 - 0.5z^{-1})(1 - z^{-1})}.$$

Using partial fraction expansion,

$$\frac{1 + z^{-1}}{(1 - 0.5z^{-1})(1 - z^{-1})} = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - z^{-1}}.$$

Multiplying both sides by the denominator,

$$1 + z^{-1} = A(1 - z^{-1}) + B(1 - 0.5z^{-1}).$$

Setting

$$z^{-1} = 1,$$

we obtain

$$2 = 0.5B$$

and therefore

$$B = 4.$$

Setting

$$z^{-1} = 2,$$

we obtain

$$3 = -A$$

and therefore

$$A = -3.$$

Thus,

$$Y(z) = \frac{-3}{1 - 0.5z^{-1}} + \frac{4}{1 - z^{-1}}.$$

Taking the inverse Z-transform,

$$y[n] = -3(0.5)^n u[n] + 4u[n].$$

Therefore,

$$\boxed{y[n] = [4 - 3(0.5)^n] u[n]}$$

4 Question 4: Sampling and Aliasing

The given continuous-time signal is

$$x(t) = \cos(2\pi(2 \text{ kHz})t) + 0.5 \cos(2\pi(5 \text{ kHz})t) .$$

The two frequency components are

$$f_1 = 2 \text{ kHz}$$

and

$$f_2 = 5 \text{ kHz}.$$

4.1 (a) Minimum Sampling Frequency

The highest frequency present in the signal is

$$f_{\max} = 5 \text{ kHz}.$$

According to the Nyquist sampling theorem,

$$f_s \geq 2f_{\max}.$$

Therefore,

$$f_s \geq 2(5 \text{ kHz})$$

and hence the Nyquist rate is

$$\boxed{f_s = 10 \text{ kHz}}.$$

Thus, a sampling frequency of at least 10 kHz is required to avoid aliasing under the ideal Nyquist condition.

4.2 (b) Apparent Frequencies at $f_s = 6 \text{ kHz}$

The sampling frequency is

$$f_s = 6 \text{ kHz}.$$

The corresponding Nyquist frequency is

$$f_N = \frac{f_s}{2} = \frac{6}{2} = 3 \text{ kHz}.$$

4.2.1 2 kHz Component

For the first component,

$$f_1 = 2 \text{ kHz}.$$

Since

$$2 \text{ kHz} < 3 \text{ kHz},$$

it lies below the Nyquist frequency and does not alias.
Therefore,

$$\boxed{f_{1,\text{apparent}} = 2 \text{ kHz}}.$$

4.2.2 5 kHz Component

For the second component,

$$f_2 = 5 \text{ kHz}.$$

Since

$$5 \text{ kHz} > 3 \text{ kHz},$$

the component aliases.
The apparent frequency is

$$f_{2,\text{apparent}} = |f_2 - f_s|.$$

Therefore,

$$\begin{aligned} f_{2,\text{apparent}} &= |5 - 6| \text{ kHz} \\ &= 1 \text{ kHz}. \end{aligned}$$

Hence,

$$\boxed{f_{2,\text{apparent}} = 1 \text{ kHz}}.$$

Therefore, the sampled signal contains apparent frequency components at

$$\boxed{2 \text{ kHz} \quad \text{and} \quad 1 \text{ kHz}}.$$

4.3 (c) Aliasing and Anti-Aliasing Filter

Aliasing occurs when a continuous-time signal is sampled at a frequency that is too low to represent its highest frequency component uniquely.

When a frequency component exceeds the Nyquist frequency

$$\frac{f_s}{2},$$

it is folded into a lower frequency and appears as a different frequency in the sampled signal.

Once aliasing has occurred, the original frequency component cannot be uniquely recovered from the sampled data.

An anti-aliasing filter is an analog low-pass filter placed before the analog-to-digital converter. It attenuates frequency components above the desired bandwidth, particularly those above

$$\frac{f_s}{2}.$$

Thus, the anti-aliasing filter reduces high-frequency components before sampling and helps prevent aliasing.

5 Question 5: Z-Transform and System Analysis

The given discrete-time signal is

$$x[n] = (0.5)^n u[n].$$

5.1 (a) Z-Transform and ROC

The Z-transform of a discrete-time signal is

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}.$$

Since

$$x[n] = (0.5)^n u[n],$$

we have

$$X(z) = \sum_{n=0}^{\infty} (0.5)^n z^{-n}.$$

This can be written as

$$X(z) = \sum_{n=0}^{\infty} (0.5z^{-1})^n.$$

Using the geometric-series relation

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}, \quad |r| < 1,$$

we obtain

$$X(z) = \frac{1}{1-0.5z^{-1}}.$$

Multiplying the numerator and denominator by z ,

$$X(z) = \frac{z}{z-0.5}.$$

For convergence,

$$|0.5z^{-1}| < 1.$$

Therefore,

$$|z| > 0.5.$$

Hence,

$$X(z) = \frac{1}{1 - 0.5z^{-1}} = \frac{z}{z - 0.5}$$

with ROC

$$|z| > 0.5.$$

5.2 (b) Inverse Z-Transform

Given

$$X(z) = \frac{z}{z - 0.8}$$

with

$$|z| > 0.8.$$

Rewrite the expression as

$$X(z) = \frac{1}{1 - 0.8z^{-1}}.$$

The ROC

$$|z| > 0.8$$

is outside the pole and corresponds to a right-sided sequence.

Using the standard Z-transform pair

$$\frac{1}{1 - az^{-1}} \longleftrightarrow a^n u[n],$$

we obtain

$$x[n] = (0.8)^n u[n].$$

5.3 (c) Stability Analysis

The system function is

$$H(z) = \frac{1}{1 - 0.6z^{-1}}$$

with ROC

$$|z| > 0.6.$$

The pole is located at

$$z = 0.6.$$

For a discrete-time LTI system, BIBO stability requires that the ROC of the system function include the unit circle

$$|z| = 1.$$

The specified ROC is

$$|z| > 0.6.$$

Since

$$1 > 0.6,$$

the unit circle lies inside the ROC.

Therefore,

The system is stable.

The reason is that the ROC includes the unit circle.

6 Summary of Final Answers

Table 2: Summary of Results

Question	Final Answer
Q1(a)(i)	$x[n] = \{2, -1, \underset{\uparrow}{3}, 4, -2\}$
Q1(a)(ii)	$x[n-2] = \{\underset{\uparrow}{2}, -1, 3, 4, -2\}$
Q1(a)(iii)	$x[-n] = \{-2, 4, \underset{\uparrow}{3}, -1, 2\}$
Q1(a)(iv)	$x[2-n] = \{-2, 4, \underset{\uparrow}{3}, -1, 2\}$
Q1(b)	Linear, time-variant, causal, unstable
Q2	$y[n] = \{\underset{\uparrow}{1}, 1, 1, 3, 2\}$
Q3(a)	$h[n] = (0.5)^n u[n] + (0.5)^{n-1} u[n-1]$
Q3(b)	$y[n] = [4 - 3(0.5)^n] u[n]$
Q4(a)	Minimum sampling rate: 10 kHz
Q4(b)	Apparent frequencies: 2 kHz and 1 kHz
Q5(a)	$X(z) = \frac{1}{1-0.5z^{-1}}$, ROC: $ z > 0.5$
Q5(b)	$x[n] = (0.8)^n u[n]$
Q5(c)	Stable